

3. Test documentation

3.1 Test specification

Test ID: FR-01

Test Name: Launching the Game

Description:

As a player, I want to open the game from my device easily so I can start playing without delays.

Preconditions:

The app is installed.

Test Steps:

Tap the app icon on the mobile/tablet or open the game on the desktop.

Wait for the application to load.

Expected Result:

The game launches successfully.

The main menu is displayed.

Test ID: FR-02

Test Name: Displaying the Main Menu

Description:

As a player, I want to see the main menu immediately when I launch the game, so I know where to start.

Preconditions:

A game application has been launched.

Test Steps:

Open the game.

Observe the screen after loading.

Expected Result:

The main menu appears with buttons to choose the difficulty level.

Test ID: FR-03**Test Name: Selecting Difficulty Level****Description:**

As a player, I want to choose how hard the game is by selecting how many card pairs I want to play with.

Preconditions:

Main menu is open.

Test Steps:

Tap the "Select Level" option.

Choose from Easy (8 pairs), Medium (15 pairs), or Hard (24 pairs).

Expected Result:

The selected level loads with the correct number of card pairs.

Test ID: FR-04**Test Name: Displaying Game Board with HUD****Description:**

As a player, I want to see a clean game board and HUD so I can track my progress and interact with the game.

Preconditions:

A level has been selected.

Test Steps:

Start any level.

Wait for the game board to load.

Expected Result:

Cards appear in a grid layout.

HUD shows a timer and a move counter.

Test ID: FR-05**Test Name: Flipping Two Cards at a Time****Description:**

As a player, I want to flip only two cards at once so I don't get overwhelmed and can focus on remembering matches.

Preconditions:

A level is in progress.

Test Steps:

Flip one card.

Flip a second card.

Try to flip a third card quickly.

Expected Result:

with clicking sound

Only two cards flip at once.

A third card cannot be flipped until the current two are evaluated.

Test ID: FR-06**Test Name: Matching Card Logic****Description:**

As a player, I want the game to tell me when two flipped cards match, so I feel rewarded for remembering correctly.

Preconditions:

Two tiles are flipped.

Test Steps:

Flip two identical animal tiles.

Watch the game's response.

Expected Result:

The system recognizes the match.

Matched cards get eliminated and match sound is being played.

Test ID: FR-07**Test Name: Hiding Unmatched Cards****Description:**

As a player, if I flip two non-matching cards, I want them to automatically flip back so I can try again.

Preconditions:

A game is in progress.

Test Steps:

Flip two cards that don't match.

Wait shortly.

Expected Result:

The cards flip back to its natural state.

Test ID: FR-08**Test Name: display of time left and clicks****Description:**

As a player, I want to keep looking at the time left to complete the game and the clicks I have made to complete the level, so I can improve over time.

Preconditions:

Timer and move tracking are active.

Test Steps:

Play through a level.

Finish the level by matching all pairs.

Expected Result:

The final score appears based on time and moves.

The result screen shows this clearly.

Test ID: FR-09**Test Name: Ending the Game Automatically****Description:**

As a player, I want the game to end when I've matched all cards so I know I completed the challenge.

Preconditions:

Only one pair remains unmatched.

Test Steps:

Flip the final matching pair.

Expected Result:

A win message appears.

The board is disabled and no further interaction is allowed.

Test ID: FR-10**Test Name: Responsive Layout Across Devices****Description:**

As a player, I want the game to look good and work properly whether I'm playing on a phone or laptop.

Preconditions:

The game runs on multiple devices.

Test Steps:

Launch game on different screen sizes.

Observe the layout and interaction.

Expected Result:

Game elements scale properly.

No UI overlaps or cut-offs.

Test ID: FR-11**Test Name: Audio Feedback for Interactions****Description:**

As a player, I want the game to play sounds when I flip cards or match them, so I get instant feedback.

Preconditions:

The game is in progress.

Test Steps:

Flip a card.

Flip a matching pair.

Expected Result:

A click sound plays on every flip.

A different sound plays on successful matches.

Test ID: FR-12**Test Name: Toggle Background Music****Description:**

As a player, I want to mute or unmute the calming background music as needed to suit my environment.

Preconditions:

Game is running.

Test Steps:

Tap the mute/unmute button in the HUD.

Expected Result:

Background music toggles on or off.

Sound effects remain active.

Test ID: FR- 13**Test Name: Touch Input Support****Description:**

As a player who cannot use a mouse yet, I want to flip cards just by tapping the screen

Preconditions:

The game is installed on a touch-supported device.

Test Steps:

Launch the game on a touch device.

Tap buttons in the main menu.

Tap two cards during gameplay.

Tap HUD elements (e.g., mute, menu).

Expected Result:

All taps are correctly registered.

Cards flip on tap.

Menu and HUD elements respond as expected.

Test ID: NFR-01**Requirement: Simple and accessible UI for children aged 3–7****Description:**

Verify that the UI is age-appropriate, with large buttons, minimal text, and engaging visuals.

Preconditions:

Game is installed and functional.

Test Steps:

Launch the game on a laptop or mobile.

Observe the main menu, game board, and button layout.

Expected Result:

Buttons are large and clearly tappable.

No long instructions or reading is required.

Navigation is intuitive and consistent across scenes.

3.2 Test protocol

- Devices used: Android phone (model), PC (OS, specs), smartboard
- Methodology: Manual testing + playtesting by kids
- Test environment: Godot editor, exported APK for desktop and Android
- Test frequency: After each feature is implemented